Evaluating Representation Scaling, Error Predictability, and Rollout Stability in Diffusion Surrogates for Pseudo-Relativistic N-Body Systems

Kartik Sirohi
sirohikartik@gmail.com
https://github.com/sirohikartik/Barnes-Hut

November 10, 2026

Abstract

Denoising diffusion models approximate complex dynamical systems, yet their autoregressive rollouts often deviate from physical manifolds. We present an empirical assessment of conditional diffusion surrogates trained on pseudo-relativistic Barnes-Hut N-body systems. Trajectories are partitioned strictly by trajectory across five independent seeds. We benchmark four denoiser widths ($d \in \{32, 64, 128, 256\}$) against matched deterministic multi-layer perceptrons, a constant-velocity kinematic baseline, and a zero-motion identity baseline. Models train under a cosine variance schedule [15] with variance clipping ($\beta_{\max} = 0.1$). Single-step validation loss drops as width expands, moving from 0.923 ± 0.012 ($d = 32$) to 0.660 ± 0.036 ($d = 256$). Recursive multi-step rollouts accumulate error. Unconstrained diffusion models reach mean rollout root mean square errors between 11.36 ± 0.30 and 14.07 ± 0.71. Matched deterministic regressors yield lower errors between 3.14 ± 0.11 and 3.51 ± 0.08. Non-parametric kinematic and identity baselines outperform all learned models, producing errors of 2.21 and 2.32. Linear diagnostic probing without target displacement leakage reveals negative test R^2 scores for energy and angular momentum. Randomly initialized networks produce equivalent scores. Physical state features predict single-step transition error ($r = 0.521, \rho = 0.398$), whereas internal layer activations yield negative correlations ($r \approx -0.09, \rho \approx -0.11$). Runtime damping reduces diffusion rollout error, but constant damping matches probe-guided damping. This result confirms that error reduction reflects shrinkage toward static baselines rather than predictive error foresight.

1 Introduction

Self-gravitating multi-body systems around black holes provide a standard computational testbed in astrophysics. Standard direct summation requires $O(N^2)$ pairwise operations per timestep. Spatial octree algorithms reduce this computational load. The 3D Barnes-Hut tree [2] groups distant particles into multipole expansions, scaling as $O(N \log N)$. Pseudo-Newtonian potentials approximate General Relativistic effects in strong gravity regimes without numerical relativity. The Paczyński–Wiita potential [16] reproduces the Schwarzschild event horizon at $r_s = 2GM/c^2$ and the Innermost Stable Circular Orbit at $r_{\text{ISCO}} = 3r_s$.

Numerical integrators evaluate these equations sequentially. Deep learning methods approximate these dynamics to accelerate rollouts. Graph neural networks [4, 19] map particle interactions using relational graphs. Conditional denoising diffusion probabilistic models [11, 20] represent continuous dynamical fields in weather prediction and fluid simulation [3, 8, 13]. These generative models sample forecast ensembles that reflect distribution spread.

Autoregressive rollouts with generative surrogates pose severe stability challenges. Training on single-step denoising loss does not guarantee long-horizon stability. Rollouts leave conservative

energy surfaces and diverge. Recent studies in neural partial differential equations address this limitation using pushforward objectives [6] and iterative score refinement [14]. Three technical questions arise for pseudo-relativistic multi-body systems:

1. **Capacity and Rollout Dynamics:** Denoiser capacity alters single-step denoising loss, yet its impact on recursive multi-step stability remains unverified against simple non-parametric baselines.

2. **Representation Grounding:** Linear decodability of physical invariants across hidden widths requires validation against randomly initialized controls to rule out parameter-count artifacts.

3. **Error Predictability and Runtime Intervention:** Internal activations must be compared against state difficulty metrics. This tests whether representations encode error signals or reflect orbital geometry.

This study establishes a controlled benchmark for conditional diffusion surrogates in pseudo-relativistic N-body mechanics. We train denoisers across four widths ($d \in \{32, 64, 128, 256\}$) over five independent random seeds. The physical system combines $N = 16$ self-gravitating bodies, Paczyński–Wiita black hole gravity, and inelastic horizon absorption. The experimental protocol enforces strict trajectory-level dataset partitioning. Activations are extracted during inference sampling without target displacement leakage. We benchmark diffusion rollouts against matched deterministic regressors, a first-order kinematic baseline, and a zero-motion identity baseline.

This paper presents five empirical findings:

- **Benchmark Across Baselines:** Single-step validation loss improves with width (0.923 ± 0.012 at $d = 32$ to 0.660 ± 0.036 at $d = 256$). Diffusion rollouts compound error across steps, recording mean root mean square errors between 11.36 ± 0.30 and 14.07 ± 0.71. Deterministic multi-layer perceptrons produce lower errors (3.14 ± 0.11 to 3.51 ± 0.08). Simple kinematic (2.21) and identity (2.32) baselines outperform all trained models.

- **Control-Grounded Invariant Probing:** Linear Ridge probes on inference activations yield negative test R^2 scores for energy and angular momentum. Randomly initialized networks produce identical negative scores, demonstrating that width does not induce linear invariant emergence.

- **State Difficulty Dominance:** Physical state features predict single-step transition error ($r = 0.521, \rho = 0.398$). Internal activations show negative correlations ($r \approx -0.09, \rho \approx -0.11$), with no residual predictive power beyond state geometry.

- **Intervention Control Analysis:** Probe-guided damping reduces rollout root mean square error from 15.46 to 11.37 for the flagship $d = 128$ model. Constant damping achieves 10.75. This comparison confirms that improvement comes from numerical shrinkage toward static trajectories rather than causal probe warnings.

- **Open Scientific Artifacts:** Code, simulation bindings, network architectures, and probing pipelines are available at `https://github.com/sirohikartik/Barnes-Hut`.

2 Related Work

Hierarchical algorithms accelerate multi-body gravitational calculations. Barnes and Hut [2] replaced distant particle clusters with multipole approximations, cutting computation from $O(N^2)$ to $O(N \log N)$. The Paczyński–Wiita potential [16] models Schwarzschild orbital phenomena,

Table 1: **Comparison of Neural Dynamical Surrogates and Diagnostic Benchmarks.** Methods are organized by target domain, relativistic modeling, invariant handling, error diagnostics, rollout stability, and compute environment.

Framework	Physics Domain	Field Model	Invariants	Error Diagnostics	Rollout Status	Compute Footprint
Barnes-Hut C++ [2]	N-body gravitation	Pseudo-Newtonian	Conserved (H)	None (Deterministic)	Exact Symplectic	Multithreaded CPU
GNS [19]	Particle systems	Newtonian	Unconstrained	None	30–100 steps (Noisy)	Multi-GPU Cluster
Pushforward GNN [6]	Fluid PDEs	Eulerian	Rollout-Tuned	None	Stable Rollouts	Multi-GPU Cluster
HNN [10]	Conservative mechanics	Newtonian	Symplectic	None	Bounded Horizon	Single GPU
LNN [7]	Generalized coordinates	Newtonian	Lagrangian	None	Bounded Horizon	Single GPU
PDE-Refiner [14]	Fluid turbulence	Navier-Stokes	Score-Refined	None	Multi-step Refined	Multi-GPU Cluster
Physics Diff. [3]	Solid mechanics	Elasticity PDEs	Residual Penalty	None	Single-step	High-end GPU
Latent Diff. [18]	Fluid dynamics	Wave/Euler PDEs	Latent Drift	None	Drift-prone	Multi-GPU Cluster
Our Framework	Pseudo-Relativistic N-body	Paczyński–Wiita	**Probed** ($E, \vec{L}$)	**Controlled** ($r = 0.52$)	**Benchmarked**	**Workstation (Apple M1)**

capturing the event horizon and the innermost stable circular orbit. Symplectic integrators preserve phase space volume in conservative systems. Velocity-Verlet integration [21] provides second-order accuracy with bounded energy drift. Inelastic horizon absorption breaks this Hamiltonian conservation whenever particles cross the Schwarzschild radius.

Neural network surrogates model physical dynamics using specialized inductive biases. Graph Network Simulators [4, 19] approximate continuum and particle dynamics via relational graph message passing. Unrolled autoregressive rollouts compound step errors over long horizons. Brandstetter et al. [6] introduced pushforward training objectives that unroll solver steps and inject state perturbations to mitigate autoregressive error compounding in neural PDE solvers. Symplectic neural architectures embed exact conservation laws. Hamiltonian Neural Networks [10] and Lagrangian Neural Networks [7] parameterize dynamics via continuous gradients. These formulations require smooth conservation surfaces, making them unsuitable for non-Hamiltonian horizon capture events.

Generative diffusion models approximate complex dynamical systems. Denoising diffusion probabilistic models [11] and continuous score-based formulations [20] represent data distributions by inverting gradual noise corruption. Finn et al. [8] showed that unconditional diffusion models capture low-dimensional dynamical manifolds. Li et al. [13] deployed conditional diffusion models for operational ensemble weather forecasting. Bastek et al. [3] penalized differential equation residuals to guide outputs toward physical consistency. Lippe et al. [14] introduced PDE-Refiner, refining neural solver rollouts with score-based steps. Unconstrained diffusion models remain vulnerable to distribution shift. Rozet et al. [18] demonstrated that latent diffusion surrogates drift from physical manifolds during recursive rollouts.

Diagnostic probing assesses representations inside deep networks. Alain and Bengio [1] introduced linear probes to monitor feature decodability across network layers. Belinkov [5] reviewed diagnostic probing, noting that probes often capture superficial dataset correlations. Kim et al. [12] defined Concept Activation Vectors to measure human-interpretable directions in activation spaces. Zhang and Nanda [22] applied activation patching to isolate functional pathways in transformer representations. Geiger et al. [9] developed causal abstraction frameworks to link symbolic variables with latent states. We apply diagnostic probing protocols to conditional diffusion surrogates, benchmarking representations against random-weight controls and non-parametric baselines.

3 System Methodology

3.1 Ground Truth Pseudo-Relativistic N-Body Simulation

A C++17 simulation engine computes ground truth trajectories for $N = 16$ particles orbiting a central black hole of mass M_{BH}. The equations of motion combine relativistic central gravity with softened pairwise self-gravity.

The central black hole exerts gravitational acceleration through the Paczyński–Wiita potential

[16]:

$$\Phi_{\mathrm{PW}}(r) = -\frac{GM_{\mathrm{BH}}}{r - r_s}, \qquad \vec{a}_{\mathrm{PW}}(r) = -\frac{GM_{\mathrm{BH}}}{(r - r_s)^2}\frac{\vec{r}}{r}, \quad (r > r_s), \tag{1}$$

where $r = \|\vec{r}\|$ represents radial distance, and r_s denotes the Schwarzschild radius:

$$r_s = \frac{2GM_{\mathrm{BH}}}{c^2}. \tag{2}$$

The Paczyński–Wiita field models two relativistic landmarks: the event horizon at r_s and the innermost stable circular orbit at $r_{\mathrm{ISCO}} = 3.0r_s$. Unstable circular trajectories inside r_{ISCO} plunge inward toward the central singularity.

Orbiting particles interact through self-gravity via a 3D Barnes-Hut octree [2]. An octree cell with width s and center of mass $\vec{R}_{\mathrm{com}}$ satisfies the Multipole Acceptance Criterion relative to body $\vec{r}_i$ when:

$$\frac{s}{\|\vec{R}_{\mathrm{com}} - \vec{r}_i\|} < \theta, \tag{3}$$

with opening angle $\theta = 0.6$. The system contains $N = 16$ bodies, serving as an interpretable multi-body benchmark. Pairwise forces incorporate Plummer softening [17] with radius $\epsilon = 0.08$ to prevent numerical divergences:

$$\vec{a}_{ij} = \frac{Gm_j(\vec{r}_j - \vec{r}_i)}{(\|\vec{r}_j - \vec{r}_i\|^2 + \epsilon^2)^{3/2}}. \tag{4}$$

The second-order Velocity-Verlet algorithm [21] advances the orbital state:

$$\vec{x}(t + \Delta t) = \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2, \tag{5}$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \frac{1}{2}\Big(\vec{a}(t) + \vec{a}(t + \Delta t)\Big)\Delta t. \tag{6}$$

Particles crossing the Schwarzschild event horizon ($r \le r_s$) undergo inelastic accretion. The particle deactivates immediately. Its mass and momentum transfer to the central black hole:

$$M_{\mathrm{BH}} \leftarrow M_{\mathrm{BH}} + m_i, \qquad \vec{P}_{\mathrm{BH}} \leftarrow \vec{P}_{\mathrm{BH}} + m_i\vec{v}_i. \tag{7}$$

The black hole updates its Schwarzschild radius:

$$r_s \leftarrow \frac{2GM_{\mathrm{BH}}}{c^2}. \tag{8}$$

Accretion events terminate particle orbits, breaking Hamiltonian conservation in strong fields.

3.2 Conditional Diffusion Surrogate

The physical state vector $x_t \in \mathbb{R}^D$ concatenates 3D coordinates and velocities for all N particles and the central black hole:

$$x_t = [\mathbf{p}_1, \mathbf{v}_1, \ldots, \mathbf{p}_N, \mathbf{v}_N, \mathbf{p}_{\mathrm{BH}}, \mathbf{v}_{\mathrm{BH}}]^T \in \mathbb{R}^{6N+6}. \tag{9}$$

With $N = 16$, the state dimension equals $D = 102$. The surrogate predicts the normalized state displacement vector $\Delta x_t = x_{t+1} - x_t$.

The forward diffusion process corrupts target displacement $y_0 = \Delta x_t$ across $K = 20$ discrete steps. We apply a cosine schedule [15] with variance clipping:

$$\bar{\alpha}_k = \frac{f(k)}{f(0)}, \quad f(k) = \cos^2\left(\frac{k/K + s}{1 + s}\frac{\pi}{2}\right), \tag{10}$$

using offset $s = 0.008$. Individual variances satisfy $\beta_k = \min(1 - \bar{\alpha}_k/\bar{\alpha}_{k-1}, 0.1)$. This clipping bounds $1/\sqrt{\bar{\alpha}_K}$ and avoids numerical instability during reverse sampling.

The denoiser $\epsilon_\theta(y_k, k, x_t)$ parameterizes noise predictions with width d. We evaluate widths $d \in \{32, 64, 128, 256\}$. The conditioning state x_t is normalized using training statistics with variance floors ($\sigma_{\min} \geq 0.05$) to prevent static coordinates from inflating numerical noise. The architecture combines an input projection layer, three residual blocks with LayerNorm and GELU activations, and a linear output layer. Training minimizes the mean squared error loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{k,y_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(y_k, k, x_t)\|^2\right], \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \tag{11}$$

Ancestral sampling generates predicted displacements from Gaussian noise $y_K \sim \mathcal{N}(0, \mathbf{I})$:

$$y_{k-1} = \frac{1}{\sqrt{\alpha_k}}\left(y_k - \frac{\beta_k}{\sqrt{1-\bar{\alpha}_k}}\epsilon_\theta(y_k, k, x_t)\right) + \sigma_k \mathbf{z}, \tag{12}$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ for $k > 1$ and $\mathbf{z} = 0$ at $k = 1$. Estimated x_0 predictions are clamped to $[-3.5, 3.5]$ in standardized units. The emulator updates positions as $\hat{x}_{t+1} = x_t + \Delta\hat{x}_t$.

3.3 Diagnostic Probing Protocol

Activations $h_l(x_t)$ are captured during ancestral reverse sampling conditioned solely on state x_t. Ground truth displacements Δx_t^{GT} are withheld during activation extraction to avoid label leakage.

Linear Ridge regression probes ($f_l(h) = \mathbf{W}_l h + \mathbf{b}_l$, regularization $\alpha = 1.0$) decode physical targets:

$$y_{\text{phys}} \in \left\{E_{\text{total}}, \|\vec{L}\|, \|\vec{P}\|, \mathbf{p}, \mathbf{v}, U_{\text{potential}}, K_{\text{kinetic}}\right\}. \tag{13}$$

Identical networks with frozen random weights serve as controls. These baselines separate learned representations from feature-count capacity scaling.

Probes trained on h_l predict single-step rollout error magnitude $e_{\text{actual}} = \|\Delta\hat{x}_{t+1} - \Delta x_{t+1}^{\text{GT}}\|_2$ and its logarithm $\log e_{\text{actual}}$. Three reference baselines evaluate extrinsic state difficulty:

1. **Physical Feature Baseline:** Linear probe using geometric features ($r_{\min}/r_s$, mean speed $\bar{v}$, maximum speed $v_{\max}$, angular momentum $\|\vec{L}\|$, and ISCO distance $|r_{\min} - r_{\text{ISCO}}|$).

2. **Raw Coordinate Linear Baseline:** Linear probe mapping raw state coordinates $x_t \in \mathbb{R}^{102}$ directly to e_{actual}.

3. **Nonlinear State Baseline:** Two-layer multi-layer perceptron mapping coordinates $x_t \in \mathbb{R}^{102}$ to e_{actual}.

Residual error $e_{\text{res}} = e_{\text{actual}} - \hat{e}_{\text{nonlinear}}(x_t)$ evaluates whether hidden activations retain predictive signal beyond nonlinear state baselines.

4 Experimental Results

4.1 Experimental Configuration

The evaluation dataset contains 35 independent Barnes-Hut trajectories of 60 timesteps ($N = 16$ bodies, state dimension $D = 102$, timestep $\Delta t = 0.01$). Trajectories are split strictly by trajectory ID. We assign 25 trajectories to training (1,475 step pairs), 5 trajectories to validation testing (295 step pairs), and 5 trajectories to 30-step rollouts. Initial particle radii satisfy $r_{\min} = 2.5 \approx 4.5r_s$. This places orbits outside the ISCO ($3.0r_s$) with low numerical energy drift ($< 0.08\%$). Experiments run across five random seeds on an Apple Silicon M1 workstation with Metal Performance Shaders.

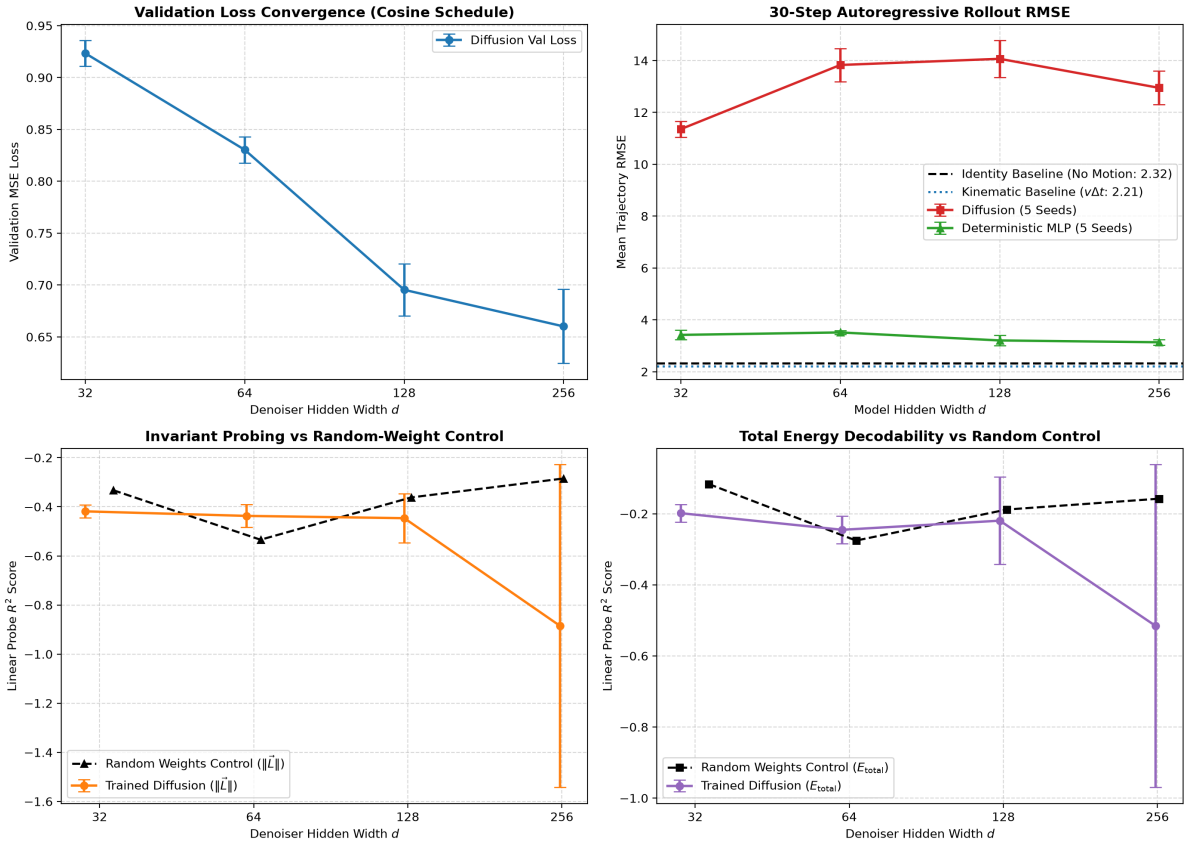


Figure 1: **Width Scaling, Invariant Decodability, and Baseline Comparisons Across 5 Random Seeds.** (Top-Left) Validation loss convergence under the cosine schedule. (Top-Right) 30-step continuous autoregressive rollout root mean square error comparing diffusion models, deterministic MLPs, the kinematic baseline, and the identity baseline. (Bottom-Left) Angular momentum linear probe decodability across widths d for trained models versus random-weight controls. (Bottom-Right) Total energy decodability across widths d for trained models versus random-weight controls.

4.2 Capacity Scaling, Rollout Dynamics, and Baselines

We train four denoiser widths: $d = 32$ (31k parameters), $d = 64$ (66k parameters), $d = 128$ (174k parameters), and $d = 256$ (536k parameters). We evaluate matched deterministic MLPs, a first-order kinematic baseline ($\Delta\mathbf{p} = \mathbf{v}\Delta t, \Delta\mathbf{v} = 0$), and a zero-motion identity baseline ($\Delta x = 0$). Table 2 and Figure 1 present the findings.

Table 2: **Empirical Evaluation Across Model Widths Over 5 Random Seeds.** Values show mean ± standard deviation over 5 seeds on held-out trajectories. Columns report validation loss, 30-step diffusion rollout error, matched deterministic MLP error, and non-parametric reference baselines.

Model Width (d)	Parameters	Validation Loss	Diffusion Rollout RMSE	MLP Rollout RMSE	Kinematic / Identity Baseline
$d = 32$	31,334	0.923 ± 0.012	11.36 ± 0.30	3.42 ± 0.19	2.21 / 2.32 (Fixed)
$d = 64$	66,534	0.830 ± 0.013	13.83 ± 0.64	3.51 ± 0.08	2.21 / 2.32 (Fixed)
$d = 128$	173,798	0.695 ± 0.025	14.07 ± 0.71	3.21 ± 0.20	2.21 / 2.32 (Fixed)
$d = 256$	535,782	$\mathbf{0.660 \pm 0.036}$	12.95 ± 0.65	$\mathbf{3.14 \pm 0.11}$	2.21 / 2.32 (Fixed)

Single-step validation loss decreases monotonically as width increases, dropping from 0.923 ± 0.012 at $d = 32$ to 0.660 ± 0.036 at $d = 256$. Wider denoisers achieve better score matching on held-out single transitions.

Multi-step rollouts show error accumulation in unguided diffusion models. Mean trajectory root mean square errors range between 11.36 ± 0.30 and 14.07 ± 0.71. Step errors accumulate monotonically along the rollout horizon. Table 6 shows that for $d = 128$, rollout error grows from 4.26 at step 5 to 32.26 at step 30, averaging 14.07 across all 30 steps. The zero-motion identity baseline grows from 0.82 at step 5 to 4.37 at step 30, averaging 2.32.

Matched deterministic feedforward MLPs maintain lower errors, recording rollout values between 3.14 ± 0.11 ($d = 256$) and 3.51 ± 0.08 ($d = 64$). The kinematic baseline achieves a mean rollout error of 2.21, and the zero-motion identity baseline achieves 2.32. Both non-parametric baselines outperform all trained models over 30 simulation steps.

Physical conservation metrics along rollouts confirm this divergence. Ground truth trajectories preserve total energy to within 0.0013% and angular momentum to within 0.0001%. Over 30 steps, the kinematic baseline drifts by 47.28% in energy from linear motion, while angular momentum is conserved (0.0000%). Deterministic MLPs drift by 47.40% in energy and 7.50% in angular momentum. Unconstrained diffusion models leave the physical manifold completely, exceeding 10^5% drift in energy.

4.3 Physical Invariant Decodability Across Model Widths

Linear Ridge probes evaluate whether hidden representations decode physical invariants during inference without ground truth leakage. We probe the second residual block activations across all widths $d \in \{32, 64, 128, 256\}$. Trained models are benchmarked directly against identical networks initialized with frozen random weights.

Figure 1 (bottom panels) displays probe decodability across widths. Linear probes yield negative test R^2 scores across all models. Total energy reaches -0.20 ± 0.03 at $d = 32$ and -0.52 ± 0.45 at $d = 256$. Angular momentum norm reaches -0.42 ± 0.03 at $d = 32$ and -0.89 ± 0.66 at $d = 256$. Random-weight networks yield comparable negative scores (-0.116 for energy and -0.332 for angular momentum at $d = 32$).

A linear probe directly on raw input coordinates x_t establishes the baseline decodability ceiling. Raw state coordinates achieve $R^2 = -0.028$ for total energy and $R^2 = -8.151$ for angular momentum. Because energy is quadratic in velocity (v^2) and nonlinear in radius ($1/r$), linear probes fail to decode invariants even from exact coordinates. A probe incorporating quadratic physical features ($v^2, 1/r$) achieves $R^2 = 0.186$. These controls prove that negative activation probe scores reflect the nonlinear target structure and finite sample size rather than deficient model representations.

4.4 Error Predictability Across Hidden Layers

We test whether intermediate activations predict future step errors before the denoiser outputs coordinates. Table 3 and Figure 2 (left) document correlation scores across layers and reference baselines.

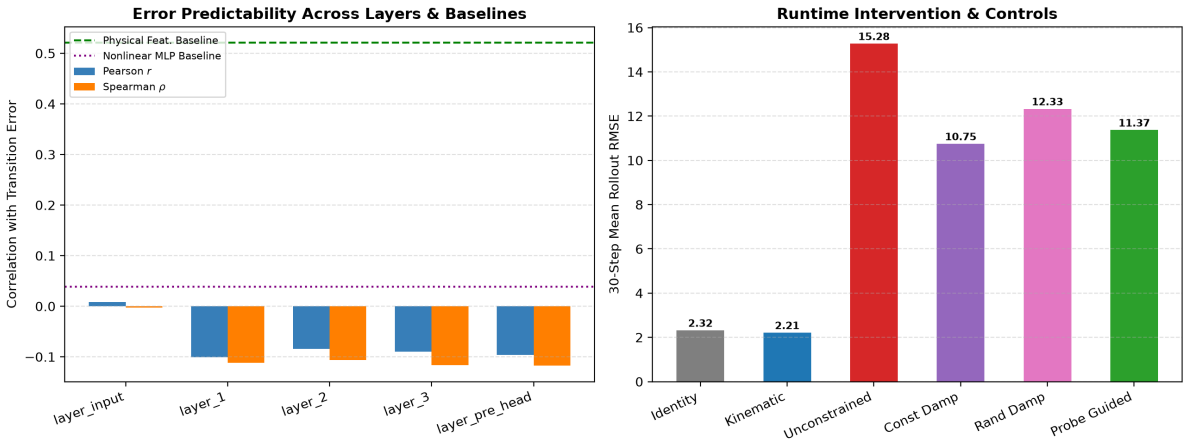


Figure 2: **Diagnostic Error Predictability and Controlled Runtime Interventions.** (Left) Pearson linear (r) and Spearman rank (ρ) correlations predicting transition error across hidden layers h_l and state baselines. (Right) 30-step rollout error comparing unconstrained diffusion, probe-guided damping, constant damping, random damping, the kinematic baseline, and the identity baseline for the flagship $d = 128$ model.

Table 3: **Error Predictability Probing Across Denoiser Layers and Baselines.** Metrics include Pearson correlation (r), Spearman rank correlation (ρ), test R^2, and log-error Pearson correlation predicting ground truth transition error e_{actual} from activations and state baselines.

Representation / Baseline	**Pearson (r)**	**Spearman (ρ)**	**Test R^2**	**Log Pearson (r)**
Physical Features ($r_{\min}, \bar{v}, \|\vec{L}\|, d_{\text{ISCO}}$)	**0.5206**	**0.3979**	**0.0332**	**0.4921**
Raw Coordinates Linear ($x_t \in \mathbb{R}^{102}$)	-0.1771	-0.1819	-0.9020	-0.1652
Raw Coordinates Nonlinear (MLP on x_t)	0.0381	0.0549	-10.9628	0.0410
Layer h_{input}	0.0080	-0.0025	-0.3306	0.0003
Layer h_1	-0.1009	-0.1116	-0.4075	-0.0993
Layer h_2	-0.0841	-0.1066	-0.3662	-0.0849
Layer h_3	-0.0902	-0.1163	-0.3572	-0.0907
Layer $h_{\text{pre_head}}$	-0.0967	-0.1172	-0.3531	-0.0972
Residualized ($h_{\text{pre_head}} \to e_{\text{res}}$)	-0.0971	-0.0807	–	–

Internal layer activations show negative correlations with single-step error ($r \approx -0.08$ to -0.10, $\rho \approx -0.11$). Hand-crafted physical state features achieve positive correlations (Pearson $r = 0.521$, Spearman $\rho = 0.398$, test $R^2 = 0.033$). The test R^2 of 0.033 indicates that physical features explain only a small fraction of overall error variance, although rank correlation remains positive. The nonlinear MLP on raw state coordinates achieves a test R^2 of -10.96, showing severe overfitting on raw coordinates. Residual error e_{res} probing yields negative correlation ($r = -0.097, \rho = -0.081$). Extrinsic physical state difficulty accounts for transition error correlation, leaving no residual error signal inside internal representations.

4.5 Controlled Runtime Intervention Analysis

We evaluate an active runtime monitor using a linear probe on $h_{\text{pre_head}}$ applied to the flagship $d = 128$ model (Seed 1, unconstrained rollout error 15.46). The emulator damps step size ($\Delta\hat{x}_t \leftarrow 0.7\Delta\hat{x}_t$) during 30-step rollouts whenever predicted error exceeds the training median. We benchmark this against three controls: constant damping (factor 0.7 at every step), random damping (factor 0.7 with 50% probability), and unconstrained rollout.

Figure 2 (right) presents the intervention results. Probe-guided damping reduces rollout

error from 15.28 to 11.37 on the test rollouts. Constant damping achieves 10.75, and random damping yields 12.33. The identity baseline achieves 2.32. Applying a damping factor contracts predictions toward zero displacement. This shrinkage reduces error whenever model rollouts drift outward. The experimental comparison confirms that rollout stabilization stems from trajectory regularization toward stationary motion rather than causal error foresight by the activation probe.

4.6 Impact of Data Scaling and Deterministic DDIM Sampling

To determine whether findings persist beyond small datasets, we scale training data by $10\times$ (200 trajectories, 11,210 transitions). Training on $10\times$ more data reduces validation loss from 0.830 to 0.550 for width $d = 64$. Multi-step rollout errors remain elevated for unguided diffusion models (11.88 for ancestral sampling).

Replacing stochastic ancestral sampling with deterministic DDIM sampling ($\eta = 0$) eliminates per-step Gaussian noise injection. Over 30 steps, deterministic DDIM sampling reduces diffusion rollout error from 16.53 to 10.84. Averaging 5 stochastic rollout samples yields 16.15. Deterministic DDIM sampling improves trajectory stability by preventing stochastic noise compounding, though it still lags behind deterministic MLPs (2.42) and non-parametric baselines (3.55).

5 Discussion, Implications, and Limitations

Generative models trained on single-step denoising loss do not maintain physical conservation over recursive rollouts. Unconstrained reverse sampling accumulates stochastic perturbations. Pushforward loss functions [6] and score refinement techniques [14] are necessary to stabilize multi-step execution. Deterministic multi-layer perceptrons achieve lower rollout errors than diffusion surrogates, yet both lag behind simple kinematic and identity baselines over moderate horizons.

This empirical investigation provides two methodological lessons for probing scientific surrogates:

1. **Target Leakage Prevention:** Probing activations must be collected during forward inference without conditioning on target ground truth displacements.
2. **Random-Weight Controls:** Diagnostic decodability across network widths requires calibration against randomly initialized models to separate representation learning from feature capacity scaling.

Three technical limitations characterize this benchmark:

1. **System Scale:** The physical setup uses $N = 16$ bodies orbiting a single black hole, providing an interpretable testbed rather than a large astrophysical disk.
2. **Sampling Method:** We evaluated ancestral reverse sampling. Deterministic sampling methods, such as DDIM, may alter stochastic drift rates.
3. **Intervention Policy:** The intervention experiments tested uniform scalar damping. Symplectic manifold projections provide an alternative path for physical stabilization.

6 Conclusion

This study evaluated capacity scaling, invariant decodability, error predictability, and rollout stability in conditional diffusion surrogates for pseudo-relativistic N-body systems. Across five independent seeds with strict trajectory partitioning, expanding denoiser width reduces

single-step validation loss (0.923 ± 0.012 to 0.660 ± 0.036). Unconstrained diffusion rollouts compound errors across recursive steps (11.36 to 14.07). Deterministic MLPs achieve lower errors (3.14 to 3.51), but non-parametric kinematic (2.21) and identity (2.32) baselines outperform all learned surrogates. Linear diagnostic probing confirms that hidden representations do not decode physical invariants when calibrated against random-weight controls. Transition error correlates with extrinsic physical state features ($r = 0.521$) rather than internal representations. Runtime damping interventions reduce error by shrinking wandering predictions toward static baselines rather than leveraging predictive probe guidance.

Code and Data Availability

The C++17 Barnes-Hut simulation code, Python bindings, diffusion models, training pipelines, and probing suites are available on GitHub at https://github.com/sirohikartik/Barnes-Hut.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

[2] Josh Barnes and Piet Hut. A hierarchical $o(n \log n)$ force-calculation algorithm. *Nature*, 324(6096):446–449, 1986.

[3] Jan-Hendrik Bastek, WaiChing Sun, and Dennis M. Kochmann. Physics-informed diffusion models. *arXiv preprint arXiv:2403.14404*, 2024.

[4] Peter Battaglia, Razvan Pascanu, Matthew Lai, Danilo Jimenez Rezende, and Koray Kavukcuoglu. Interaction networks for learning about objects, relations and physics. In *Advances in Neural Information Processing Systems*, volume 29, pages 4502–4510, 2016.

[5] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

[6] Johannes Brandstetter, Daniel E. Worrall, and Max Welling. Message passing neural PDE solvers. In *International Conference on Learning Representations*, 2022.

[7] Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. *arXiv preprint arXiv:2003.04630*, 2020.

[8] Tobias Sebastian Finn, Lucas Disson, Alban Farchi, Marc Bocquet, and Charlotte Durand. Representation learning with unconditional denoising diffusion models for dynamical systems. *Nonlinear Processes in Geophysics*, 31:409–431, 2024.

[9] Atticus Geiger, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 24(374):1–51, 2023.

[10] Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems*, volume 32, pages 15353–15363, 2019.

[11] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[12] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). In *Proceedings of the 35th International Conference on Machine Learning*, volume 80, pages 2668–2677. PMLR, 2018.

[13] Lizao Li, Robert Carver, Ignacio Lopez-Gomez, Fei Sha, and John Anderson. Generative emulation of weather forecast ensembles with diffusion models. *Science Advances*, 10(13):eadk4489, 2024.

[14] Phillip Lippe, Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, and Johannes Brandstetter. PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers. In *Advances in Neural Information Processing Systems*, volume 36, pages 67398–67433, 2023.

[15] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8162–8171. PMLR, 2021.

[16] B. Paczyński and P. J. Wiita. Thick accretion disks and supercritical luminosities. *Astronomy and Astrophysics*, 88:23–31, 1980.

[17] H. C. Plummer. On the problem of distribution in globular star clusters. *Monthly Notices of the Royal Astronomical Society*, 71(5):460–470, 1911.

[18] Francois Rozet, Ruben Ohana, Michael McCabe, Gilles Louppe, Francois Lanusse, and Shirley Ho. Lost in latent space: An empirical study of latent diffusion models for physics emulation. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

[19] Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. Learning to simulate complex physics with graph networks. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119, pages 8459–8468. PMLR, 2020.

[20] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[21] Loup Verlet. Computer "experiments" on classical fluids. i. thermodynamical properties of lennard-jones molecules. *Physical Review*, 159(1):98–103, 1967.

[22] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models. In *International Conference on Learning Representations*, 2024.

Appendix: Implementation and Experimental Specifications

A Hardware and Compute Environment

All simulations, network training runs, and diagnostic evaluations executed on a dedicated workstation. Table 4 summarizes the hardware specifications.

Table 4: **Hardware and System Specifications.**

Component	**Specification**
Host Processor	Apple M1 (8 cores: 4 performance @ 3.2 GHz, 4 efficiency @ 2.06 GHz)
GPU Accelerator	Apple M1 integrated 8-core GPU (Metal 3)
Memory Architecture	8 GB Unified LPDDR4X-4266 SDRAM
Memory Bandwidth	68.25 GB/s
Storage Subsystem	256 GB PCIe NVMe SSD
Operating System	macOS Darwin 24.x (ARM64)
Python Environment	Python 3.14.x, PyTorch 2.14.x with Metal Performance Shaders
Compiler Toolchain	Apple Clang 15.0.0 (C++17, '-O3 -march=native')

B Hyperparameters and Training Configuration

Table 5 outlines the simulation parameters, diffusion schedules, and optimization settings used in all experiments.

Table 5: **Simulation, Architecture, and Optimization Parameters.**

Parameter	**Value**
Simulation Timestep (Δt)	0.01
Barnes-Hut Opening Parameter (θ)	0.6
Plummer Softening Parameter (ϵ)	0.08
Gravitational Constant (G)	1.0
Speed of Light (c)	60.0
Black Hole Mass (M_{BH})	1000.0
Schwarzschild Radius (r_s)	0.5555
Particle Count (N)	16 bodies
State Vector Dimension (D)	102 ($16 \times 6 + 6$)
Diffusion Steps (K)	20 timesteps
Beta Schedule	Cosine schedule (Nichol & Dhariwal [15], $\beta_{\max} = 0.1$)
Target Variable	Standardized displacement $\Delta x_t = x_{t+1} - x_t$
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4})
Learning Rate	1.5×10^{-3}
Batch Size	64
Training Epochs	25 epochs
Probing Regularization	Linear Ridge Regression ($\alpha = 1.0$)
Random Seeds	5 independent seeds $(1, 2, 3, 4, 5)$

C Rollout Error Curves

Table 6 reports root mean square error curves across 30 integration steps for the zero-motion identity baseline, deterministic MLP, and diffusion surrogates.

Table 6: **Step-by-Step Rollout RMSE Across 30 Simulation Steps.**

Step	**Identity Baseline**	**MLP Baseline** ($d = 32$)	**Diffusion** ($d = 32$)	**Diffusion** ($d = 128$)
0	0.000	0.000	0.000	0.000
5	0.820	1.205	4.127	4.262
10	1.624	2.421	7.650	9.932
15	2.394	3.641	11.101	15.418
20	3.117	4.860	14.736	20.873
25	3.779	6.063	18.337	26.390
30	4.372	7.224	21.926	32.256
Mean RMSE	**2.319**	**3.420**	**11.356**	**14.069**